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|-----------------------------------------------------------------------------------|----------------|------------------------------------------|----------------------------------|
|  | Description    | INS PELGRAZ 6MG/0.6ML (MY) MOCK-24-00291 |                                  |
|                                                                                   | Country        | Malaysia                                 |                                  |
|                                                                                   | Language       | EN                                       |                                  |
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|                                                                                   | AW Update Date | 28/03/24,15/04/24,10/06/24, 13/01/26     |                                  |
| Prepared By<br>Regulatory Affairs                                                 |                | Checked By<br>Regulatory Affairs         | Approved By<br>Quality Assurance |
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IBPLpih-0001  
MOCK-24-00291



**Pelgraz 6 mg solution for injection in pre-filled syringe**  
pegfilgrastim



**Pelgraz 6 mg solution for injection in pre-filled syringe**  
pegfilgrastim

1. **NAME OF THE MEDICINAL PRODUCT**  
Pelgraz 6 mg solution for injection in pre-filled syringe

2. **QUALITATIVE AND QUANTITATIVE COMPOSITION**  
Each pre-filled syringe contains 6 mg of pegfilgrastim\* in 0.6 mL solution for injection. The concentration is 10 mg/mL based on protein only\*\*.

\*Produced in *Escherichia coli* cells by recombinant DNA technology followed by conjugation with polyethylene glycol (PEG).  
\*\* The concentration is 20 mg/mL if the PEG moiety is included.

The potency of this medicinal product should not be compared to the potency of another pegylated or non-pegylated protein of the same therapeutic class. For more information, see section 5.1.

**Excipients with known effect**  
Each pre-filled syringe contains 30 mg sorbitol (E420) (see section 4.4). For the full list of excipients, see section 6.1.

3. **PHARMACEUTICAL FORM**  
Solution for injection in pre-filled syringe. Clear, colourless solution for injection.

4. **CLINICAL PARTICULARS**

4.1 **Therapeutic indications**  
Reduction in the duration of neutropenia, the incidence of febrile neutropenia and the incidence of infection as manifested by febrile neutropenia in patients treated with cytotoxic chemotherapy for malignancy (with the exception of chronic myeloid leukaemia and myelodysplastic syndromes).

4.2 **Posology and method of administration**  
  
**Posology**  
Adults (≥ 18 years): One 6 mg dose (a single pre-filled syringe) of Pelgraz is recommended for each chemotherapy cycle, administered as a subcutaneous injection approximately 24 hours following cytotoxic chemotherapy.

**Method of administration**  
Pelgraz is injected subcutaneously. The injections should be given into the thigh, abdomen or upper arm. For instructions on handling of the medicinal product before administration, see section 6.6.

**Paediatric population**  
The safety and efficacy of Pelgraz in children and adolescents has not yet been established. Currently available data are described in sections 4.8, 5.1 and 5.2 but no recommendation on a posology can be made.

**Patients with renal impairment**  
No dose change is recommended in patients with renal impairment, including those with end stage renal disease.

4.3 **Contraindications**  
Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4 **Special warnings and precautions for use**  
Pelgraz is a biosimilar medicinal product of Neulastim. Pelgraz is not interchangeable or automatically substitutable with Neulastim

**Traceability**  
In order to improve the traceability of biological medicinal products, the trade name and the batch number of the administered product should be clearly recorded (or stated) in the patient file.

**Acute myeloid leukaemia (AML)**  
Limited clinical data suggest a comparable effect on time to recovery of severe neutropenia for pegfilgrastim to filgrastim in patients with *de novo* AML (see section 5.1). However, the long-term effects of pegfilgrastim have not been established in AML; therefore, it should be used with caution in this patient population.

Granulocyte-colony stimulating factor can promote growth of myeloid cells *in vitro* and similar effects may be seen on some non-myeloid cells *in vitro*.

The safety and efficacy of pegfilgrastim have not been investigated in patients with myelodysplastic syndrome, chronic myelogenous leukaemia, and in patients with secondary AML; therefore, it should not be used in such patients. Particular care should be taken to distinguish the diagnosis of blast transformation of chronic myeloid leukaemia from AML.

The safety and efficacy of pegfilgrastim administration in *de novo* AML patients aged < 55 years with cytogenetics t(15;17) have not been established.

The safety and efficacy of pegfilgrastim have not been investigated in patients receiving high dose chemotherapy. This medicinal product should not be used to increase the dose of cytotoxic chemotherapy beyond established dose regimens.

**Pulmonary adverse reactions**  
Pulmonary adverse reactions, in particular interstitial pneumonia, have been reported after G-CSF administration. Patients with a recent history of pulmonary infiltrates or pneumonia may be at higher risk (see section 4.8).  
The onset of pulmonary signs such as cough, fever, and dyspnoea in association with radiological signs of pulmonary infiltrates, and deterioration in pulmonary function along with increased neutrophil count may be preliminary signs of Adult Respiratory Distress Syndrome (ARDS). In such circumstances pegfilgrastim should be discontinued at the discretion of the physician and the appropriate treatment given (see section 4.8).

**Glomerulonephritis**  
Glomerulonephritis has been reported in patients receiving filgrastim and pegfilgrastim. Generally, events of glomerulonephritis resolved after dose reduction or withdrawal of filgrastim and pegfilgrastim. Urinalysis monitoring is recommended.

**Capillary leak syndrome**  
Capillary leak syndrome has been reported after granulocyte-colony stimulating factor administration and is characterised by hypotension, hypoalbuminaemia, oedema and

haemoconcentration. Patients who develop symptoms of capillary leak syndrome should be closely monitored and receive standard symptomatic treatment, which may include a need for intensive care (see section 4.8).

**Splenomegaly and splenic rupture**  
Generally asymptomatic cases of splenomegaly and cases of splenic rupture, including some fatal cases, have been reported following administration of pegfilgrastim (see section 4.8). Therefore, spleen size should be carefully monitored (e.g. clinical examination, ultrasound). A diagnosis of splenic rupture should be considered in patients reporting left upper abdominal pain or shoulder tip pain.

**Thrombocytopenia and anaemia**  
Treatment with pegfilgrastim alone does not preclude thrombocytopenia and anaemia because full dose myelosuppressive chemotherapy is maintained on the prescribed schedule. Regular monitoring of platelet count and haematocrit is recommended. Special care should be taken when administering single or combination chemotherapeutic medicinal products which are known to cause severe thrombocytopenia.

**Myelodysplastic syndrome and acute myeloid leukaemia in breast and lung cancer patients**  
In the post-marketing observational study setting, pegfilgrastim in conjunction with chemotherapy and/or radiotherapy has been associated with development of myelodysplastic syndrome (MDS) and acute myeloid leukaemia (AML) in breast and lung cancer patients (see section 4.8). Monitor breast and lung cancer patients for signs and symptoms of MDS/AML.

**Sickle cell anaemia**  
Sickle cell crises have been associated with the use of pegfilgrastim in patients with sickle cell trait or sickle cell disease (see section 4.8). Therefore, physicians should use caution when prescribing pegfilgrastim in patients with sickle cell trait or sickle cell disease, should monitor appropriate clinical parameters and laboratory status and be attentive to the possible association of this medicinal product with splenic enlargement and vaso-occlusive crisis.

**Leukocytosis**  
White blood cell (WBC) counts of 100 x 10<sup>9</sup>/L or greater have been observed in less than 1% of patients receiving pegfilgrastim. No adverse reactions directly attributable to this degree of leukocytosis have been reported. Such elevation in WBCs is transient, typically seen 24 to 48 hours after administration and is consistent with the pharmacodynamic effects of this medicinal product. Consistent with the clinical effects and the potential for leukocytosis, a WBC count should be performed at regular intervals during therapy. If leukocyte counts exceed 50 x 10<sup>9</sup>/L after the expected nadir, this medicinal product should be discontinued immediately.

**Hypersensitivity**  
Hypersensitivity, including anaphylactic reactions, occurring on initial or subsequent treatment have been reported in patients treated with pegfilgrastim. Permanently discontinue pegfilgrastim in patients with clinically significant hypersensitivity. Do not administer pegfilgrastim to patients with a history of hypersensitivity to pegfilgrastim or filgrastim. If a serious allergic reaction occurs, appropriate therapy should be administered, with close patient follow-up over several days.

**Immunogenicity**  
As with all therapeutic proteins, there is a potential for immunogenicity. Rates of generation of antibodies against pegfilgrastim is generally low. Binding antibodies do occur as expected with all biologics; however, they have not been associated with neutralising activity at present.

**Aortitis**  
Aortitis has been reported after filgrastim or pegfilgrastim administration in healthy subjects and in cancer patients. The symptoms experienced included fever, abdominal pain, malaise, back pain and increased inflammatory markers (e.g. C-reactive protein and WBC count). In most cases aortitis was diagnosed by CT scan and generally resolved after withdrawal of filgrastim or pegfilgrastim. See also section 4.8.

**Mobilisation of PBPC**  
The safety and efficacy of Pelgraz for the mobilisation of blood progenitor cells in patients or healthy donors has not been adequately evaluated.

**Other special precautions**  
Increased haematopoietic activity of the bone marrow in response to growth factor therapy has been associated with transient positive bone-imaging findings. This should be considered when interpreting bone-imaging results.

**Excipients with known effect**  
Sorbitol  
This medicinal product contains 50 mg sorbitol in each unit volume, which is equivalent to 30 mg per 6 mg dose.

**Sodium**  
Pelgraz contains less than 1 mmol (23 mg) sodium per 6 mg dose, that is to say essentially 'sodium-free'.

**All patients**  
The needle cover of the pre-filled syringe contains dry natural rubber (a derivative of latex), which may cause allergic reactions.

4.5 **Interaction with other medicinal products and other forms of interaction**  
Due to the potential sensitivity of rapidly dividing myeloid cells to cytotoxic chemotherapy, pegfilgrastim should be administered at least 24 hours after administration of cytotoxic chemotherapy. In clinical trials, pegfilgrastim has been safely administered 14 days before chemotherapy.  
Concomitant use of Pelgraz with any chemotherapeutic medicinal product has not been evaluated in patients. In animal models concomitant administration of pegfilgrastim and 5-fluorouracil (5-FU) or other antimetabolites has been shown to potentiate myelosuppression.

Possible interactions with other hematopoietic growth factors and cytokines have not been specifically investigated in clinical trials.

The potential for interaction with lithium, which also promotes the release of neutrophils, has not been specifically investigated. There is no evidence that such an interaction would be harmful.

The safety and efficacy of Pelgraz have not been evaluated in patients receiving

chemotherapy associated with delayed myelosuppression e.g. nitrosoureas.

Specific interaction or metabolism studies have not been performed; however, clinical trials have not indicated an interaction of pegfilgrastim with any other medicinal products.

4.6 **Fertility, pregnancy and lactation**

**Pregnancy**  
There are no or limited amount of data from the use of pegfilgrastim in pregnant women. Studies in animals have shown reproductive toxicity (see section 5.3). Pegfilgrastim is not recommended during pregnancy and in women of childbearing potential not using contraception.

**Breast-feeding**  
There is insufficient information on the excretion of pegfilgrastim/metabolites in human milk, a risk to the newborns/infants cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from pegfilgrastim therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

**Fertility**  
Pegfilgrastim did not affect reproductive performance or fertility in male or female rats at cumulative weekly doses approximately 6 to 9 times higher than the recommended human dose (based on body surface area) (see section 5.3).

4.7 **Effects on ability to drive and use machines**  
Pegfilgrastim has no or negligible influence on the ability to drive and use machines.

4.8 **Undesirable effects**  
**Summary of the safety profile**  
The most frequently reported adverse reactions were bone pain (very common (≥ 1/10)) and musculoskeletal pain (common [≥ 1/100 to < 1/10]). Bone pain was generally of mild to moderate severity, transient and could be controlled in most patients with standard analgesics.

Hypersensitivity-type reactions, including skin rash, urticaria, angioedema, dyspnoea, erythema, flushing, and hypotension occurred on initial or subsequent treatment with pegfilgrastim (uncommon [≥ 1/1,000 to < 1/100]). Serious allergic reactions, including anaphylaxis can occur in patients receiving pegfilgrastim (uncommon) (see section 4.4).

Capillary Leak Syndrome, which can be life-threatening if treatment is delayed, has been reported as uncommon (≥ 1/1,000 to < 1/100) in cancer patients undergoing chemotherapy following administration of granulocyte-colony stimulating factors; see section 4.4 and section "Description of selected adverse reactions" below.

Splenomegaly, generally asymptomatic, is uncommon.

Splenic rupture including some fatal cases is uncommonly reported following administration of pegfilgrastim (see section 4.4).

Uncommon pulmonary adverse reactions including interstitial pneumonia, pulmonary oedema, pulmonary infiltrates and pulmonary fibrosis have been reported. Uncommonly, cases have resulted in respiratory failure or ARDS, which may be fatal (see section 4.4).

Isolated cases of sickle cell crises have been reported in patients with sickle cell trait or sickle cell disease (uncommon in sickle cell patients) (see section 4.4).

**Tabulated list of adverse reactions**  
The data in the table below describe adverse reactions reported from clinical trials and spontaneous reporting. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

| MedDRA system organ class                                           | Adverse reactions     |                                                            |                                                                                                                                                                                           |                                |                        |
|---------------------------------------------------------------------|-----------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------|
|                                                                     | Very common (≥ 1/10)  | Common (≥ 1/100 to < 1/10)                                 | Uncommon (≥ 1/1,000 to < 1/100)                                                                                                                                                           | Rare (≥ 1/10,000 to < 1/1,000) | Very rare (< 1/10,000) |
| Neoplasms benign, malignant and unspecified (incl cysts and polyps) |                       |                                                            | Myelodysplastic syndrome, Acute myeloid leukaemia,                                                                                                                                        |                                |                        |
| Blood and lymphatic system disorders                                |                       | Thrombocytopenia <sup>1</sup><br>Leukocytosis <sup>1</sup> | Sickle cell crisis <sup>2</sup> ;<br>Splenomegaly <sup>2</sup> ;<br>Splenic rupture <sup>2</sup>                                                                                          |                                |                        |
| Immune system disorders                                             |                       |                                                            | Hypersensitivity reactions;<br>Anaphylaxis                                                                                                                                                |                                |                        |
| Metabolism and nutrition disorders                                  |                       |                                                            | Elevations in uric acid                                                                                                                                                                   |                                |                        |
| Nervous system disorders                                            | Headache <sup>3</sup> |                                                            |                                                                                                                                                                                           |                                |                        |
| Vascular disorders                                                  |                       |                                                            | Capillary leak Syndrome <sup>1</sup>                                                                                                                                                      | Aortitis                       |                        |
| Respiratory, thoracic and mediastinal disorders                     |                       |                                                            | Adult Respiratory Distress Syndrome <sup>2</sup> ;<br>Pulmonary adverse reactions (interstitial pneumonia, pulmonary oedema, pulmonary infiltrates and pulmonary fibrosis)<br>Haemoptysis | Pulmonary haemorrhage          |                        |

Front Side



|                                                                                   |                |                                          |
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|                                                               |                     |                                                                                                                                  |                                                                                                                                                              |  |  |
|---------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Gastrointes-<br>tinal<br>disorders                            | Nausea <sup>1</sup> |                                                                                                                                  |                                                                                                                                                              |  |  |
| Skin and<br>subcutaneous<br>tissue<br>disorders               |                     |                                                                                                                                  | Sweet's syndrome<br>(acute febrile<br>dermatosis) <sup>1,2</sup><br>Cutaneous<br>Vasculitis <sup>1,2</sup>                                                   |  |  |
| Musculoskeletal<br>and connective<br>tissue<br>disorders      | Bone<br>pain        | Musculoskeletal<br>pain (myalgia,<br>arthralgia, pain<br>in extremity,<br>back pain,<br>musculo-<br>skeletal pain,<br>neck pain) |                                                                                                                                                              |  |  |
| Renal and<br>urinary<br>disorders                             |                     |                                                                                                                                  | Glomerulone<br>phritis <sup>1</sup>                                                                                                                          |  |  |
| General<br>disorders and<br>administrative<br>site conditions |                     | Injection site<br>pain <sup>1</sup><br>Non-cardiac<br>chest pain                                                                 | Injection site<br>reactions <sup>1</sup>                                                                                                                     |  |  |
| Investigations                                                |                     |                                                                                                                                  | Elevations in<br>lactate<br>dehydrogenase<br>and alkaline<br>phosphatase <sup>1</sup> ;<br>Transient<br>elevations in<br>LFTs for<br>ALT or AST <sup>1</sup> |  |  |

<sup>1</sup> See section "Description of selected adverse reactions" below.  
<sup>2</sup> This adverse reaction was identified through post-marketing surveillance but not observed in randomised, controlled clinical trials in adults. The frequency category was estimated from a statistical calculation based upon 1,576 patients receiving pegfilgrastim in nine randomised clinical trials.

**Description of selected adverse reactions**  
Uncommon cases of Sweet's syndrome have been reported, although in some cases underlying haematological malignancies may play a role.

Uncommon events of cutaneous vasculitis have been reported in patients treated with pegfilgrastim. The mechanism of vasculitis in patients receiving pegfilgrastim is unknown.

Injection site reactions, including injection site erythaema (uncommon) as well as injection site pain (common events) have occurred on initial or subsequent treatment with pegfilgrastim.

Common cases of leukocytosis (WBC > 100 x 10<sup>9</sup>/L) have been reported (see section 4.4).

Reversible, mild to moderate elevations in uric acid and alkaline phosphatase, with no associated clinical effects, were uncommon; reversible, mild to moderate elevations in lactate dehydrogenase, with no associated clinical effects, were uncommon in patients receiving pegfilgrastim following cytotoxic chemotherapy.

Nausea and headaches were very commonly observed in patients receiving chemotherapy.

Uncommon elevations in liver function tests (LFTs) for alanine aminotransferase (ALT) or aspartate aminotransferase (AST), have been observed in patients after receiving pegfilgrastim following cytotoxic chemotherapy. These elevations are transient and return to baseline.

An increased risk of MDS/AML following treatment with Pelgraz in conjunction with chemotherapy and/or radiotherapy has been observed in an epidemiological study in breast and lung cancer patients (see section 4.4).

Common cases of thrombocytopenia have been reported.

Cases of capillary leak syndrome have been reported in the post-marketing setting with granulocyte-colony stimulating factor use. These have generally occurred in patients with advanced malignant diseases, sepsis, taking multiple chemotherapy medicinal products or undergoing apheresis (see section 4.4).

**Paediatric population**  
The experience in children is limited. A higher frequency of serious adverse reactions in younger children aged 0-5 years (92%) has been observed compared to older children aged 6-11 and 12-21 years respectively (80% and 67%) and adults. The most common adverse reaction reported was bone pain (see section 5.1 and 5.2).

**Reporting of suspected adverse reactions**  
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

**4.9 Overdose**  
Single doses of 300 mcg/kg have been administered subcutaneously to a limited number of healthy volunteers and patients with non-small cell lung cancer without serious adverse reactions. The adverse reactions were similar to those in subjects receiving lower doses of pegfilgrastim.

5. PHARMACOLOGICAL PROPERTIES

**5.1 Pharmacodynamic properties**  
Pharmacotherapeutic group: immunostimulants, colony stimulating factor; ATC Code: L03AA13

Pelgraz is a biosimilar medicinal product.

Human granulocyte colony stimulating factor (G-CSF) is a glycoprotein, which regulates the production and release of neutrophils from the bone marrow. Pegfilgrastim is a covalent conjugate of recombinant human G-CSF (r-metHuG-CSF) with a single 20 kd PEG molecule. Pegfilgrastim is a sustained duration form of filgrastim due to decreased renal clearance. Pegfilgrastim and filgrastim have been shown to have identical modes of action, causing a marked increase in peripheral blood neutrophil counts within 24 hours, with minor increases in monocytes and/or lymphocytes. Similarly, to filgrastim, neutrophils produced in response to pegfilgrastim show normal or enhanced function as demonstrated by tests of chemotactic and phagocytic function. As with other haematopoietic growth factors, G-CSF has shown *in vitro* stimulating properties on human endothelial cells. G-CSF can promote growth of myeloid cells, including malignant cells, *in vitro* and similar effects may be seen on some non-myeloid cells *in vitro*.

Information provided below is based on studies conducted with Pelgraz. A comparative, randomised, assessor-blind, single-dose, 2-way cross-over study was conducted to assess similarity of single SC administration of 3 mg (T1 and R1) and 6 mg (T2 and R2) of Pelgraz and Europe sourced Neulasta in terms of pharmacokinetics and pharmacodynamics in healthy subjects. Of the total of 344 subjects (172 subjects per dose level group) dosed, 148 subjects completed 3 mg and 151 completed 6 mg for both periods. Baseline-adjusted absolute neutrophil count (ANC) was compared for area under the effective concentration curve (AUC<sub>0-24</sub>) and maximum measured ANC (E<sub>max</sub>). Biosimilarity was concluded if the 95% CIs for the baseline-adjusted ANC E<sub>max</sub> and AUC<sub>0-24</sub> Geometric mean ratio (GMR) of T2 versus R2 lay within the 80.00% to 125.00% reference interval. Results tabulated below:

|                                                | Treatment T2 (Test) |     | Treatment R2 (Reference) |     |         |                         |                   |           |
|------------------------------------------------|---------------------|-----|--------------------------|-----|---------|-------------------------|-------------------|-----------|
| Parameter                                      | Geometric LSMs      | n   | Geometric LSMs           | n   | GMR (%) | 95% Confidence interval | Intra-subject CV% | Power (%) |
| AUC <sub>0-24</sub> (hr × 10 <sup>9</sup> /μL) | 4757.2              | 144 | 4720.0                   | 144 | 100.79  | 97.75 -103.92           | 13.17             | 100.00    |
| E <sub>max</sub> (× 10 <sup>9</sup> /μL)       | 31.20               | 148 | 31.61                    | 148 | 98.70   | 95.52 -101.98           | 14.23             | 100.00    |

Treatment T2: A single subcutaneous dose of Pelgraz Administered at 6 mg/0.6 mL (Test Product T2)  
Treatment R2: A single subcutaneous dose of Neulasta® administered at 6 mg/0.6 mL (Reference Product R2)

A phase III, randomized, active controlled, assessor-blinded study was conducted to evaluate safety and efficacy of Pelgraz versus US- and EU-licensed Neulasta® in subjects with stage IIa, IIb or IIIa breast cancer receiving TAC anticancer chemotherapy (docetaxel, doxorubicin, cyclophosphamide) in adjuvant setting. Five hundred and ninety-five (595) female subjects were randomized from 56 investigational centers in 11 countries in the study. Out of the 595 subjects, 589 were randomized and dosed. Patients were randomized to either Pelgraz or US-licensed Neulasta or EU-approved Neulasta in a 2:1:1 ratio. Fixed dose of 6 mg/0.6 mL was administered once per chemotherapy cycle for 6 cycles. The duration of severe neutropenia (DSN) in Cycle 1 was the primary endpoint for the assessment of similarity of efficacy of Pelgraz to Neulasta US and Pelgraz to Neulasta EU. For declaring equivalence, the 95% CI should lie within the range of -0.5 to 0.5 day. As pre-specified in the protocol, the primary analysis population was the full analysis set (FAS) and the per protocol (PP) analysis set was supportive. There were two groups analysed in the FAS - As randomized and As treated. The results are tabulated below:

| Statistics for duration (days) | Pelgraz      | Neulasta US  | Neulasta EU  | Pelgraz - Neulasta US | Pelgraz - Neulasta EU | Neulasta EU - Neulasta US |
|--------------------------------|--------------|--------------|--------------|-----------------------|-----------------------|---------------------------|
| FAS - As Randomized            |              |              |              |                       |                       |                           |
| LS Mean                        | 1.63         | 1.39         | 1.61         | 0.24                  | 0.02                  | 0.21                      |
| 95% CI                         | 1.47 to 1.79 | 1.17 to 1.61 | 1.38 to 1.83 | -0.03 to 0.51         | -0.25 to 0.30         | -0.10 to 0.53             |
| FAS - As Treated               |              |              |              |                       |                       |                           |
| LS Mean                        | 1.62         | 1.39         | 1.63         | 0.23                  | -0.01                 | 0.24                      |
| 95% CI                         | 1.46 to 1.77 | 1.17 to 1.61 | 1.41 to 1.86 | 0.04 to 0.50          | -0.29 to 0.26         | -0.07 to 0.56             |

In Cycle 1, Grade 4 neutropenia occurred in 77.2%, 75.0% and 79.6% subjects in the Pelgraz, Neulasta US and Neulasta EU arms, respectively, and Grade 3 neutropenia occurred in 9.5%, 13.5% and 8.8% of subjects, respectively. The majority of cases of Grade 3 and Grade 4 neutropenia occurred during the first cycle. The mean of peak ANC was reached on Day 3.4, 3.1 and 3.1 in Pelgraz, Neulasta US and Neulasta EU arms, respectively. The mean day to ANC nadir was reported as Day 7.1, 7.1 and 7.3 in the Pelgraz, Neulasta US arm and Neulasta EU arm, respectively, and mean day of recovery was reached on Day 9.4, 9.5 and 9.2, respectively. The rates of febrile neutropenia (post-hoc chi-square test: P=0.548) among the treatment arms during Cycle 1 were 5.1%, 4.1% and 3.4% in the Pelgraz, Neulasta US and Neulasta EU arms, respectively and throughout the treatment period across all cycles were 5.8%, 4.7% and 3.4% subjects respectively.

Information provided below is based on the Neulastim data. In two randomised, double-blind, pivotal studies in patients with high-risk stage II-IV breast cancer undergoing myelosuppressive chemotherapy consisting of doxorubicin and docetaxel, use of pegfilgrastim, as a single once per cycle dose, reduced the duration of neutropenia and the incidence of febrile neutropenia similarly to that observed with daily administrations of filgrastim (a median of 11 daily administrations). In the absence of growth factor support, this regimen has been reported to result in a mean duration of grade 4 neutropenia of 5 to 7 days, and a 30-40% incidence of febrile neutropenia. In one study (n = 157), which used a 6 mg fixed dose of pegfilgrastim the mean duration of grade 4 neutropenia for the pegfilgrastim group was 1.8 days compared with 1.6 days in the filgrastim group (difference 0.23 days, 95% CI -0.15, 0.63). Over the entire study, the rate of febrile neutropenia was 13% of pegfilgrastim-treated patients compared with 20% of filgrastim-treated patients (difference 7%, 95% CI of -19%, 5%). In a second study (n = 310), which used a weight-adjusted dose (100 mcg/kg), the mean duration of grade 4 neutropenia for the pegfilgrastim group was 1.7 days, compared with 1.8 days in the filgrastim group (difference 0.03 days, 95% CI -0.36, 0.30). The overall rate of febrile neutropenia was 9% of patients treated with pegfilgrastim and 18% of patients treated with filgrastim (difference 9%, 95% CI of -16.8%, -1.1%).

In a placebo-controlled, double blind study in patients with breast cancer the effect of pegfilgrastim on the incidence of febrile neutropenia was evaluated following administration of a chemotherapy regimen associated with a febrile neutropenia rate of 10-20% (docetaxel 100 mg/m<sup>2</sup> every 3 weeks for 4 cycles). Nine hundred and twenty-eight

patients were randomised to receive either a single dose of pegfilgrastim or placebo approximately 24 hours (day 2) after chemotherapy in each cycle. The incidence of febrile neutropenia was lower for patients randomised to receive pegfilgrastim compared with placebo (1% versus 17%, p<0.001). The incidence of hospitalisations and intravenous anti-infective use associated with a clinical diagnosis of febrile neutropenia was lower in the pegfilgrastim group compared with placebo (1% versus 14%, p < 0.001; and 2% versus 10%, p<0.001).

A small (n = 83), Phase II, randomised, double-blind study in patients receiving chemotherapy for *de novo* AML compared pegfilgrastim (single dose of 6 mg) with filgrastim, administered during induction chemotherapy. Median time to recovery from severe neutropenia was estimated as 22 days in both treatment groups. Long term outcome was not studied (see section 4.4).

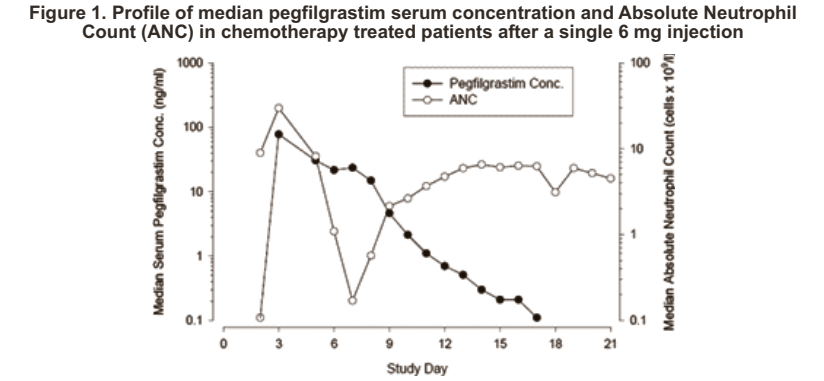
In a phase II (n = 37) multicentre, randomised, open-label study of paediatric sarcoma patients receiving 100 mcg/kg pegfilgrastim following cycle 1 of vincristine, doxorubicin and cyclophosphamide (VAdriaC/IE) chemotherapy, a longer duration of severe neutropenia (neutrophils < 0.5 x 10<sup>9</sup>) was observed in younger children aged 0-5 years (8.9 days) compared to older children aged 6-11 years and 12-21 years (6 days and 3.7 days, respectively) and adults. Additionally a higher incidence of febrile neutropenia was observed in younger children aged 0-5 years (75%) compared to older children aged 6-11 years and 12-21 years (70% and 33%, respectively) and adults (see sections 4.8 and 5.2).

**5.2 Pharmacokinetic properties**  
Information provided below is based on studies conducted with Pelgraz. The pharmacokinetics of Pelgraz was studied in a comparative, randomized, single-dose, assessor-blinded, 2-way crossover study. A total of 344 subjects (172 subjects per dose level group) were dosed in the study according to the randomization scheme. Of these 344 subjects, 299 subjects completed both periods of the study (148 completed in the 3 mg dose group and 151 completed in the 6 mg dose group). Of 299 subjects who completed both the study periods, 232 were included in the pharmacokinetics summary statistics and statistical comparisons (144 subjects for 3 mg Pelgraz vs. EU-Neulasta and 148 for 6 mg Pelgraz vs. EU-Neulasta). Biosimilarity was concluded if the 90% CIs for the pegfilgrastim C<sub>max</sub>, AUC<sub>0-24</sub>, and AUC<sub>0-∞</sub> GMRs of T2 versus R2 (6 mg/0.6 mL dose level) lay within the 80.00% to 125.00% reference interval. Results are tabulated below:

|                                  | Treatment T2 (Test) |     | Treatment R2 (Reference) |     |         |                |                   |           |
|----------------------------------|---------------------|-----|--------------------------|-----|---------|----------------|-------------------|-----------|
| Parameter                        | Geometric LSMs      | n   | Geometric LSMs           | n   | GMR (%) | 95% CI         | Intra-subject CV% | Power (%) |
| AUC <sub>0-24</sub> (ng × hr/mL) | 16276               | 146 | 15406                    | 146 | 105.65  | 99.60 - 112.06 | 31.10             | 99.87     |
| AUC <sub>0-∞</sub> (ng × hr/mL)  | 16330               | 142 | 15446                    | 142 | 105.72  | 99.55 - 112.28 | 31.34             | 99.82     |
| C <sub>max</sub> (ng/mL)         | 394.4               | 148 | 380.6                    | 148 | 103.62  | 98.19 - 109.35 | 28.53             | 100.00    |

Treatment T2: A single subcutaneous dose of Pelgraz administered at 6 mg/0.6 mL (Test Product T2)  
Treatment R2: A single subcutaneous dose of Neulasta® administered at 6 mg/0.6 mL (Reference Product R2)

Information provided below is based on the Neulastim data. After a single subcutaneous dose of pegfilgrastim, the peak serum concentration of pegfilgrastim occurs at 16 to 120 hours after dosing and serum concentrations of pegfilgrastim are maintained during the period of neutropenia after myelosuppressive chemotherapy. The elimination of pegfilgrastim is non-linear with respect to dose; serum clearance of pegfilgrastim decreases with increasing dose. Pegfilgrastim appears to be mainly eliminated by neutrophil mediated clearance, which becomes saturated at higher doses. Consistent with a self-regulating clearance mechanism, the serum concentration of pegfilgrastim declines rapidly at the onset of neutrophil recovery (see figure 1).



Due to the neutrophil-mediated clearance mechanism, the pharmacokinetics of pegfilgrastim is not expected to be affected by renal or hepatic impairment. In an open-label, single dose study (n = 31) various stages of renal impairment, including end-stage renal disease, had no impact on the pharmacokinetics of pegfilgrastim.

**Elderly**  
Limited data indicate that the pharmacokinetics of pegfilgrastim in elderly subjects (> 65 years) is similar to that in adults.

**Paediatric population**  
The pharmacokinetics of pegfilgrastim were studied in 37 paediatric patients with sarcoma, who received 100 mcg/kg pegfilgrastim after the completion of VAdriaC/IE chemotherapy. The youngest age group (0-5 years) had a higher mean exposure to pegfilgrastim (AUC) (± Standard Deviation) (47.9 ± 22.5 mcg·hr/mL) than older children aged 6-11 years and 12-21 years (22.0 ± 13.1 mcg·hr/mL and 29.3 ± 23.2 mcg·hr/mL, respectively) (see section 5.1). With the exception of the youngest age group (0-5 years), the mean AUC in paediatric subjects appeared similar to that for adult patients with high-risk stage II-IV breast cancer and receiving 100 mcg/kg pegfilgrastim after the completion of doxorubicin/docetaxel (see sections 4.8 and 5.1).

**5.3 Preclinical safety data**  
Information provided below is based on the Neulastim data.

Preclinical data from conventional studies of repeated dose toxicity revealed the expected pharmacological effects including increases in leukocyte count, myeloid hyperplasia in bone marrow, extramedullary haematopoiesis and splenic enlargement.

There were no adverse effects observed in offspring from pregnant rats given pegfilgrastim subcutaneously, but in rabbits pegfilgrastim has been shown to cause embryo/fetal toxicity (embryo loss) at cumulative doses approximately 4 times the recommended human dose, which were not seen when pregnant rabbits were exposed to the recommended human dose. In rat studies, it was shown that pegfilgrastim may cross the placenta. Studies in rats indicated that reproductive performance, fertility, oestrous cycling, days between pairing and coitus, and intrauterine survival were unaffected by pegfilgrastim given subcutaneously. The relevance of these findings for humans is not known.

6. PHARMACEUTICAL PARTICULARS

**6.1 List of excipients**  
Sodium acetate\*  
Sorbitol (E420)  
Polysorbate 20  
Water for injections

\*Sodium acetate is formed by titrating glacial acetic acid with sodium hydroxide.

**6.2 Incompatibilities**  
This medicinal product must not be mixed with other medicinal products, particularly with sodium chloride solutions.

**6.3 Shelf life**  
3 years.

**6.4 Special precautions for storage**  
Store in a refrigerator (2 °C – 8 °C). Pelgraz may be exposed to room temperature (not above 25°C ± 2°C) for a maximum single period of up to 72 hours. Pelgraz left at room temperature for more than 72 hours should be discarded. Do not freeze. Accidental exposure to freezing temperatures for a single period of less than 24 hours does not adversely affect the stability of Pelgraz. Keep the container in the outer carton in order to protect from light.

**6.5 Nature and contents of container**  
Pre-filled syringe (type I glass) with a permanently attached stainless steel injection needle, with needle safety guard, Safe'n'Sound® Passive Safety System includes a needle guard, plunger rod and extended finger flange which helps preventing needle stick injuries. The extended finger flange (EFF) is designed to improve the grip and the comfort of the body flanges and in general the handling of the device. In addition, it allows visualization of low fill volume inside the syringe. The needle cover of the pre-filled syringe contains dry natural rubber (see section 4.4). Each pre-filled syringe contains 0.6 mL of solution for injection. Pack size of one pre-filled syringe with one alcohol swab, in a blistered packaging.

**6.6 Special precautions for disposal and other handling**  
Before use, Pelgraz solution should be inspected visually for particulate matter. Only a solution that is clear and colourless should be injected. Excessive shaking may aggregate pegfilgrastim, rendering it biologically inactive. Allow the pre-filled syringe to reach room temperature before injecting. Using the pre-filled syringe with a needle safety guard. The needle safety guard covers the needle after injection to prevent needle stick injury. This does not affect normal operation of the syringe. Depress the plunger rod and push firmly at the end of the injection to ensure that syringe emptying is completed. Hold the skin securely until the injection is completed. Keep the syringe still and slowly lift your thumb from the plunger rod head. The plunger rod will move up with your thumb and the spring retracts the needle from the site, into the Needle safety guards.

**Disposal**  
Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

**7. MANUFACTURED BY**  
Intas Pharmaceuticals Limited, Biopharma Division  
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**8. PRODUCT REGISTRATION HOLDER**  
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**9. DATE OF REVISION OF THE TEXT**  
Jan 2026